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The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Abstract:

Introduction

Diabetes mellitus continues to be a major contributor to morbidity and mortality worldwide. Despite considerable effort, outcomes continue to be sub-optimal, even in resource-rich countries. Studies show optimal glycaemic control reduces morbidity, but remains elusive in a significant portion of patients with diabetes. Research shows that the use of continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) clinically benefits selected subsets of patients with diabetes in specialty practices; however the majority of patients with diabetes are cared for by the primary care workforce. Although studies have evaluated the use of CGM/FGM in specialty settings, we found no systematic reviews for evaluating the use of CGM/FGM in primary care.

Objectives

This systematic review aims to answer the questions: “Is there a role for the use of CGM or FGM in the primary care of patients with diabetes mellitus?” and “Who are the primary care patients with diabetes who are most likely to benefit from the use of CGM/FGM?”

Methods and analysis

This protocol aligns with the PRISMA-P guidelines. A systematic search will be conducted in PubMed, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science without any geographical or publication year restrictions. Filtering for English language only will be applied. We will include studies investigating CGM/FGM use and reporting primary and secondary outcome measures of glycaemic control, including haemoglobin A1c, time in target range (TIR), time below range (TBR), time above range (TAR) and hypoglycaemia, and patient and physician satisfaction with CGM/FGM use. We will examine which patient populations appear to benefit from CGM/FGM. The Covidence systematic review tool will support the blinded screening and study selection process. Screening and selection of articles will be conducted by two researchers. Cochrane's ROBINS-I and II tools will be used where appropriate to assess the risk of bias.

Ethics and dissemination: The systematic review methodology does not require ethics approval due to the nature of the study design. It is anticipated that the study findings will be relevant across disciplines and to a wide readership. The results of the

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systematic review will be published in a peer-reviewed journal and will be publicly available.

Prospero registration number: Submitted for peer-review on Jan 11, 2021. Receipt number 229416.

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Strengths and limitations of this study:

- The novelty of this systematic review will be the investigation of whether the same clinical benefits of CGM/FGM use in the specialty setting will be seen in the primary care setting as well, resulting in recommendations of use in primary care.
- This systematic review will evaluate if there is any benefit (including better glycaemic control, reduced hypoglycaemia, and improved patient and physician satisfaction) to using CGM/FGM in the primary care setting.
- We will specify which patient populations of persons with diabetes appear to benefit from CGM/FGM.
- Limitations include a language bias as only databases in the English language will be searched.
- Inclusion of non-peer reviewed, grey literature in the review may reduce the strength of evidence.

Introduction:

Diabetes mellitus continues to be a major contributor to morbidity and mortality worldwide. Despite considerable expenditures of precious healthcare funds, outcomes continue to be quite poor, even in resource-rich countries with developed healthcare systems. Diabetes and impaired fasting glucose are strongly associated with a large increase in cardiovascular and all-cause mortality.[1,2] Optimal glycaemic control has been shown in numerous studies to reduce the risk of morbidity,[3,4] but it has proved to be elusive in a significant portion of the patient population suffering from diabetes.[5]

Continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) are technologies developed to add further precision to the monitoring and management of diabetes. Both CGM and FGM involve the use of subcutaneous sensors that sample interstitial glucose levels in the patient. In CGM, the sensor automatically sends glucose values to a receiver as often as every 5 minutes,[6] and it can be used to alert the patient when the glucose trend is expected to exceed set limits, i.e., result in hypoglycaemia or

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hyperglycaemia. Fingerstick calibration is required at least daily with CGM.[7] FGM, also called intermittently scanned CGM (isCGM), uses similar approaches, but values are measured on demand, i.e., the patient uses their smartphone or other device to interrogate the sensor to measure the current interstitial glucose level, rather than having continuous measurements made. FGM does not require fingerstick calibration, as calibration is performed at the factory.[8]

Numerous studies and reviews have evaluated the use of CGM and FGM in caring for a selected subset of patients with diabetes, and they have shown some clinical benefit with regards to the secondary endpoint of goal directed glycaemic control.[9] This benefit has been demonstrated in both patients with either type 1[10-13] or type 2 diabetes.[14,15] There is also evidence that the use of CGM in the management of pregnant patients with type 1 diabetes results in improved neonatal outcomes,[16] and CGM may improve the glycaemic control of adolescents with type 1 diabetes.[17] Although there is no recognized standard for what constitutes good glycaemic control, and most guidelines are now encouraging individualized hemoglobin A1c goals based on patients' age, comorbidities, and life expectancy, for the purposes of this review we will use the American Diabetes Association guidelines for a goal A1c in most patients of less than 7% (ADA Guidelines 2019).[18]

A major limiting factor in achieving glucose targets with current standard practice is hypoglycaemia with intensification of medical management.[19] CGM/FGM have been shown to reduce episodes of both daytime and night-time hypoglycaemia.[15,20,21] Although definitions of hypoglycaemia vary, for the purposes of this paper, blood glucose less than 70 mg/dL will be considered hypoglycaemia. Interestingly, a systematic review found that hypoglycaemia was very common in older adults treated with glucose-lowering agents, occurring in 28-65% of participants in the included studies, most of whom (80-100%) were asymptomatic.[22] Hypoglycaemia in this older patient population is associated with worsening cognitive impairment, dementia,[23] falls, fractures, and cardiovascular complications, including death.[24] Finally, recurrent hypoglycaemia can lead to impaired awareness of subsequent hypoglycaemic

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episodes.[25] Although the use of CGM/FGM was not found in a systematic review to lead to restoration of awareness of hypoglycaemia, the review found that CGM can reduce rates of severe hypoglycaemia without worsening glycaemic control.[26]

Studies have highlighted that only consistent use of CGM yields improved clinical outcomes,[27-29] which suggests patient satisfaction is clinically relevant. Although cutaneous complications with the use of CGM or FGM occur, less pain or discomfort was reported by most users of CGM/FGM than with the use of self-monitoring blood glucoses to test capillary blood glucose[30] resulting in a high overall level of satisfaction with CGM.[31]

Surveys have shown that there are an insufficient number of endocrinologists to care for the large population of patients with diabetes, and in fact, the US primary care workforce cares for 85% of patients with diabetes according to data from a 2014 survey.[32] However, although numerous studies have evaluated the use of CGM/FGM in specialty settings, our search showed there are no published systematic reviews for evaluating the use of CGM in the primary care setting. Thus, the purpose of this systematic review is to determine the benefits of CGM/FGM in the primary care of diabetes mellitus.

Objectives:

This systematic review aims to answer the following questions: Do the findings of improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction reported in the specialty setting with addition of CGM or FGM translate to the primary care setting? If so, what subgroups of primary care patients with diabetes are most likely to benefit?

Methods and analysis:

This protocol uses the PRISMA-P guidelines (Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocol)[33] and is submitted for registration in the

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international prospective register of systematic reviews, PROSPERO: receipt number 229416. Amendments to this protocol will be updated in PROSPERO, if necessary. It is anticipated that the study will commence in February 2021 to December 2021. A copy of the PRISMA-P checklist is available in the online supplementary materials file 1.

Eligibility criteria

The PICO (population, intervention, comparison, outcomes) required by the PRISMA format are as follows:

Population

Inclusion criteria are RCTs or observational studies with patients of any age with type 1 diabetes, type 2 diabetes or gestational diabetes under the care of a primary care provider, including those co-managed with an endocrinologist. We will exclude studies which were primarily focused on participants who were critically ill, hospitalized or comprised of patients using closed loop systems.

Intervention

We will evaluate primary studies that report outcomes of the use of CGM or FGM in patients with diabetes in the primary care setting.

Comparison

The review seeks to compare the outcomes of the included studies on CGM/FGM with the practice of using hemoglobin A1c with or without self-monitoring of blood glucose levels in persons with diabetes in primary care practices.

Outcomes

We have defined the primary outcome as the meeting of the American Diabetes Association goals for glycaemic control in most patients of a hemoglobin A1c of 7% or less. We have included secondary outcomes of hospitalization, change from baseline A1c, time in target range (TIR), time above target range (TAR), and time

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below target range (TBR),[34] and frequency of hypoglycaemia (<=3.8 mmol/L, or 68 mg/dL) and severe hypoglycaemia (< 3.0 mmol/L, or <=54 mg/dL). For the secondary outcomes of TIR, TAR, and TBR, we used a recent consensus report for patients with type 1 diabetes produced by the joint effort of multiple well recognized organizations that defined the target range of 3.9 - 10.0 mmol (70.2-180 mg/dL).[35] The goal of TIR > 70% for both patients with type 1 and type 2 diabetes (>50% for "older/high risk" individuals defined as cognitive deficits, renal disease, joint disease, osteoporosis, fracture, cardiovascular disease, and/or those requiring assisted care) was defined by a different recent consensus report.[36] For the subset of pregnant patients with diabetes, we will use the more stringent ADA guidelines.[37] If data exist, we will report the patient-oriented clinically significant outcomes of patient satisfaction and diabetes related complications such as major cardiovascular events, neuropathy, nephropathy, and retinopathy.

Type of studies

We will include studies published in the English language that evaluate the use of CGM or FGM in a primary care clinic regardless of the design of the study, with a minimum of 20 participants. We will exclude editorials, case reports, conference abstracts, and comments.

Information sources

We will include the electronic databases PubMed, MEDLINE, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science in the search. Grey literature will be included for review. All databases will be searched from their inception to February 2021.

Search strategy

A comprehensive, systematic search strategy will be constructed by a medical librarian (LÖ) and reviewed by the subject specialists (AK, RDG & JK). Search terms will be defined based on the PICO (population, intervention, comparator and outcomes) and the study inclusion and exclusion criteria as outlined above. We will use PubMed and

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PubMed's MeSH in the pre-search phase to systematically identify relevant search term variations and search techniques. The search will be performed without any filters or geographical or publication year restrictions. We will use the "Text Word" field (or similar) for the best possible information retrieval. Hand screening of the references lists in the included studies will in addition be conducted.

The result from the database search and details of the de-duplication of records will be documented in a PRISMA flow-diagram and a full search log with notes, results and details for all searchers will be presented in the review. The pre-search developed in PubMed in December-January 2021 can be found in the online supplementary materials file 2. The strategy developed in PubMed will be adopted and applied in all selected databases in February 2021. A search update in all databases will be conducted before finalizing the manuscript to ensure the inclusion of eventual relevant studies published after the search in February 2021.

Study records

Data management

We will upload all records identified in the database search to Covidence systematic review software[38] for automatic de-duplication and blinded screening, conflict resolving, study selection, and data extraction. The screening and extraction process in Covidence is blinded. Studies identified to be included in the review will be exported to EndNote, a citation management tool.

Selection process

The selection process in Covidence will follow the PRISMA workflow. Two independent reviewers (JK & RDG) will initially screen the title and abstracts of all identified papers based on the pre-set inclusion and exclusion criteria. A third reviewer (AK) will resolve eventual conflicts identified by the software. Full text of the papers identified as potentially eligible in the title and abstract screening will be reviewed independently by JK and AK. Reason for exclusion will be documented

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based on the pre-set study exclusion criteria imported to Covidence. RDG will resolve eventual conflicting exclusion reasons or paper inclusion/exclusion with the help of the software. Only the studies fulfilling the inclusion criteria will be fully screened for data extraction. The result from the screening and selection process will be documented in a PRISMA flow-diagram.

Data collection process

Two independent reviewers (JK & AK) will extract data from the final studies identified eligible to be included in the review. The Covidence Extraction module will be used to match the objectives of this study. Data to be extracted will include basic characteristics of the publication such as authors, year of publication, journal and country. Additionally, the study design, population, duration and CGM usage duration will be taken into consideration. The primary variables to be collected will be age; type of diabetes (type 1, type 2 or gestational); the use of CGM, FGM or both; episodes of hypoglycemia as defined above in the Outcomes section of the PICO statement; the number of insulin injections given per day (0, 1 or multiple); hospitalization for complications of either diabetes or diabetes-related treatment; patient satisfaction scores and physician satisfaction scores. Quantitative data to be extracted include the weighted mean difference of change from baseline hemoglobin A1c, time in (TIR), above (TAR), and below (TBR) the goal range after CGM/FGM is utilized. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be extracted.

Eventual discrepancies between the reviewers will be addressed with a third reviewer (RDG) and discussed until consensus is reached. Authors of the included papers will be contacted by e-mail in-case additional data is needed.

Risk of bias in individual studies

Two independent reviewers (JK & RDG) will evaluate the risk of bias and the quality of evidence of all included studies with the help of the systematic review software Covidence. The assessment of the selected studies will be conducted using a tool according to the type of study (e.g., ROBINS I for non-randomized studies and ROBINS II for RCTs). The reviewers will

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critically assess each study for its methodology to determine if any attrition, publication, outcome, or conclusion biases are present. An appraisal of the quality of each study will be given. Eventual disagreements, unusual patterns or discrepancies in the assessment will be reviewed and discussed together with a third reviewer (AK) until consensus is reached. The existence of systematic differences in baseline characteristics between the groups compared in a study will be addressed in the discussion at the end of the study.

Data synthesis

We will summarize the PICO characteristics of each study together with the intervention and other variables stated above. Thereafter we will determine which studies have similar characteristics and outcomes and this will allow us to group them. A qualitative narrative will be formulated for each subgroup of persons with diabetes treated in the primary care setting. These subgroups are anticipated to include type 1 diabetes, type 2 diabetes without insulin therapy, type 2 diabetes with only long-acting insulin therapy, type 2 diabetes with multiple daily insulin injections and gestational diabetes. The qualitative narrative for each of these subgroups will include basic characteristics of the publication such as journal, country, authors and year of publication. Additionally, the study design, population, duration and CGM usage duration will be taken into consideration. Quantitative data to be reported for each study include the weighted mean difference of change from baseline hemoglobin A1c, time in (TIR), above (TAR), and below (TBR) the goal range after CGM/FGM is utilized. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be reported with their respective 95% CIs, along with qualitative data on patient and physician satisfaction. Finally, we will give an appraisal of the utility of the addition of CGM or FGM in the treatment of that subgroup, and in our discussion, analyze which subgroups of patients with diabetes mellitus appear to benefit most from CGM/FGM. A pilot study will be conducted by JK & RDG to confirm the final design of the data collection table ahead of the final extraction and to assess the feasibility of the data synthesis.

Strength of evidence

We will use the GRADE (Grading of Recommendations Assessment, Development and Evaluation) tool to determine the strength of evidence, including publication bias, indirectness, inconsistencies, imprecision and risk of bias. Two independent reviewers (JK & RDG) will

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evaluate the quality of each study, and a third independent reviewer (AK) will resolve any potential conflict.

Patient and public involvement: No patient or members of the public will be included in the performance of this study.

Discussion

This systematic review will be conducted to compare the benefits of adding CGM/FGM to usual care in the monitoring and management of diabetes mellitus in primary care. This promising technology has shown multiple benefits in the specialty care setting, but it is unclear whether or not these benefits would be realized in the primary care setting. This review will allow us to determine whether the addition of CGM/FGM will lead to better outcomes in achieving glycemic control while avoiding frequent hypoglycemia, both of which have been strongly associated with reduced morbidity and mortality.

Ethics and dissemination: Considering the study design of systematic reviews, ethics approval is not needed. We intend to publish the review in a peer-reviewed journal where the results will be available in an open access format.

Authors' contributions: AK is the primary author for the study. The study design and draft of the protocol was conducted by all authors. JK is the corresponding author. LÖ developed the search strategy and provided feedback on the study protocol design. The final manuscript was approved by all authors.

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Competing interest: None declared

Patient consent for publication: Not required.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist

Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and hemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol.

Authors: Dr. Alexander Kieu, Dr. Romona Govender Devi, Ms. Linda Östlundh & Dr. Jeffrey King.

Manuscript submission date: 2021-01-XX

Section and topic	Item No	Checklist item	Item reported on page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	**1**
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	**N/A**
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	**2 and 4** (Submitted on Jan. 11, 2021. Receipt number: 229416)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	**1**
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	**7**
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	**N/A**
Support:			
Sources	5a	Indicate sources of financial or other support for the review	**7**
Sponsor	5b	Provide name for the review funder and/or sponsor	**7**
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	**7**
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	**2-4**

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Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	**4 -5**
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	**4-5**
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	**5**
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, Including planned limits, such that it could be repeated	**5 and supplementary materials file 2**
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	**5**
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) hrough each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	**6**
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	**6**
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources) any pre-planned data assumptions and simplifications	**6**
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	**4-5**
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	**6**
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	**6**
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned explorationof consistency (such as I^2, Kendall's τ)	**6**
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses,	

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		meta-regression)	**6**
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	**N/A**
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	**6**
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	**7**

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary materials file 2: preliminary search strategy

Source: PubMed
Search date: 2021-01-24
Search specifications: All search terms are searched in the search field “TextWord” and in MeSH (when available). A filter for English language is applied.
Result: 110 records

Preliminary search strategy:

(("primary healthcare"[Text Word] OR "primary health care"[Text Word] OR “primary health-care”[Text Word] OR "Primary Health Care"[Mesh] OR "primary care"[Text Word] OR "primary medical care"[Text Word] OR “family medicine” [Text Word] OR "Family Practice"[Mesh] OR "family practice"[Text Word] OR “family physician*”[Text Word] OR “family practitioner*”[Text Word] OR “family clinic*”[Text Word] OR “general practice” [Text Word] OR “general practitioner*”[Text Word] OR "General Practitioners"[Mesh] OR "general clinic*"[Text Word] OR "general physician*"[Text Word] OR “ambulatory care” [Text Word] OR “ambulatory health care”[Text Word] OR “ambulatory clinic*” [Text Word] OR "ambulatory physician*”[Text Word] OR “ambulatory practitioner*”[Text Word] OR “ambulatory health center*”[Text Word] OR "Ambulatory Care Facilities"[Mesh] OR "Prenatal Care"[Mesh] OR "prenatal care"[Text Word] OR "prenatal health care"[Text Word] OR “antenatal care”[Text Word] OR “pregnancy care” [Text Word] OR “maternity care” [Text Word] OR “antenatal health care”[Text Word] OR “pregnancy health care” [Text Word] OR “maternity health care” [Text Word]) AND ("CGM"[Text Word] OR "Continuous Glucose Monitor*"[Text Word] OR "flash glucose monitor*" [Text Word]))

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The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Secondary Subject Heading:	General practice / Family practice
Keywords:	PRIMARY CARE, Diabetes in pregnancy < DIABETES & ENDOCRINOLOGY, Diabetes & endocrinology < INTERNAL MEDICINE, General diabetes < DIABETES & ENDOCRINOLOGY, GENERAL MEDICINE (see Internal Medicine), Paediatric endocrinology < PAEDIATRICS

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Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Keywords: Continuous glucose monitoring; primary care; diabetes mellitus; glycaemic control; systematic review; protocol

Word Count: 2,553 words

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Abstract

Introduction:

Diabetes mellitus continues to be a major contributor to morbidity and mortality worldwide. Studies demonstrate that optimal glycaemic control reduces morbidity but remains elusive in a significant portion of patients with diabetes. Research shows that the use of continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) clinically benefits selected subsets of patients with diabetes in specialty practices; however, the majority of patients with diabetes are cared for by the primary care workforce. Although studies have evaluated the use of CGM/FGM in specialty settings, we found no systematic reviews for evaluating the use of CGM/FGM in primary care.

This systematic review aims to answer the questions: "Does the use of CGM/FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

Methods and Analysis:

Aligning with the PRISMA-P guidelines, a search will be conducted in PubMed, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science. We will include studies investigating CGM/FGM use and reporting the primary outcome measure of haemoglobin A1c and secondary outcome measures of hypoglycaemia, TIR, TBR, TAR, and patient/staff satisfaction. We will examine which patient populations appear to benefit from

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CGM/FGM. Three independent researchers will use the Covidence systematic review software for blinded screening and study selection. The NHLBI quality assessment tool and GRADE will be used to assess the risk of bias and quality of evidence.

Ethics and dissemination:

The systematic review methodology does not require ethics approval due to the nature of the study design. Study findings will be publicly available to a wide readership across disciplines and will be published in a peer-reviewed journal.

Prospero registration number: CRD42021229416

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Strengths and limitations of this study:

- To our knowledge, this will be the first systematic review to analyze the clinical benefits of CGM and FGM in the primary care setting.
- The evidence is located through a comprehensive and systematic search in seven biomedical and nursing databases.
- The Covidence systematic review software will be used for blinded screening, conflict resolving, extraction and quality assessment by three independent reviewers.
- NHLBI and GRADE will be used to assess the risk of bias and quality of evidence.
- Limitations include an English language bias and exclusion of grey, non-peer reviewed studies.

Introduction:

Diabetes mellitus continues to be a major contributor to morbidity and mortality worldwide. Despite considerable expenditures of precious healthcare funds, outcomes continue to be quite poor, even in resource-rich countries with developed healthcare systems. Diabetes and impaired fasting glucose are strongly associated with a large increase in cardiovascular and all-cause mortality.[1,2] Optimal glycaemic control has been shown in numerous studies to reduce the risk of morbidity,[3,4] but it has proved to be elusive in a significant portion of the patient population suffering from diabetes.[5]

Continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) are technologies developed to add further precision to the monitoring and management of diabetes. Both CGM and FGM involve the use of subcutaneous sensors that sample interstitial glucose levels in the

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patient. In CGM, the sensor automatically sends glucose values to a receiver as often as every 5 minutes,[6] and it can be used to alert the patient when the glucose trend is expected to exceed set limits, i.e., result in hypoglycaemia or hyperglycaemia. Fingerstick calibration is required at least daily with CGM.[7] FGM, also called intermittently scanned CGM (isCGM), uses similar approaches, but values are measured on demand, i.e., the patient uses their smartphone or other device to interrogate the sensor to measure the current interstitial glucose level, rather than having continuous measurements made. FGM does not require fingerstick calibration, as calibration is performed at the factory.[8]

Numerous studies and reviews have evaluated the use of CGM and FGM in caring for a selected subset of patients with diabetes, and they have shown some clinical benefit with regards to the secondary endpoint of goal directed glycaemic control.[9] This benefit has been demonstrated in both patients with either type 1[10-13] or type 2 diabetes.[14,15] There is also evidence that the use of CGM in the management of pregnant patients with type 1 diabetes results in improved neonatal outcomes,[16] and CGM may improve the glycaemic control of adolescents with type 1 diabetes.[17] Although there is no recognized standard for what constitutes good glycaemic control, and most guidelines are now encouraging individualized haemoglobin A1c (HbA1c) goals based on patients' age, comorbidities, and life expectancy, for the purposes of this review we will use the American Diabetes Association (ADA) guidelines 2019 for a goal A1c in most patients of less than 7% (53 mmol/mol, 154 mg/dL, 8.6 mmol/L).[18]

A major limiting factor in achieving glucose targets with current standard practice is hypoglycaemia with intensification of medical management.[19] CGM/FGM have been shown to

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reduce episodes of both daytime and night-time hypoglycaemia.[15,20,21] Interestingly, a systematic review found that hypoglycaemia was very common in older adults treated with glucose-lowering agents, occurring in 28-65% of participants in the included studies, most of whom (80-100%) were asymptomatic.[22] Hypoglycaemia in this older patient population is associated with worsening cognitive impairment, dementia,[23] falls, fractures, and cardiovascular complications, including death.[24] Finally, recurrent hypoglycaemia can lead to impaired awareness of subsequent hypoglycaemic episodes.[25] Although the use of CGM/FGM was not found in a systematic review to lead to restoration of awareness of hypoglycaemia, the review found that CGM can reduce rates of severe hypoglycaemia without worsening glycaemic control.[26]

Studies have highlighted that only consistent use of CGM yields improved clinical outcomes,[27-29] which suggests patient satisfaction is clinically relevant. Although cutaneous complications with the use of CGM or FGM occur, less pain or discomfort was reported by most users of CGM/FGM than with the use of self-monitoring blood glucoses to test capillary blood glucose[30] resulting in a high overall level of satisfaction with CGM.[31]

Surveys have shown that there are an insufficient number of endocrinologists to care for the large population of patients with diabetes, and in fact, the US primary care workforce cares for 85% of patients with diabetes according to data from a 2014 survey.[32] However, although numerous studies have evaluated the use of CGM/FGM in specialty settings, our search showed there are no published systematic reviews for evaluating the use of CGM in the primary care setting. Thus, the purpose of this systematic review is to determine the benefits of CGM/FGM in the primary care of diabetes mellitus.

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Objectives:

This systematic review aims to answer the following questions: Does the use of CGM/FGM in the primary care of patients with diabetes result in improved glycaemic control, decreased rates of hypoglycaemia, and improved patient and physician satisfaction? If so, what subgroups of primary care patients with diabetes are most likely to benefit?

Methods and analysis:

This protocol follows the PRISMA-P guidelines (Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocol)[33] and registered in the international prospective register of systematic reviews, PROSPERO: registration number CRD42021229416. Eventual amendments to this protocol will be recorded in the PROSPERO registration. It is anticipated that the study will commence in February 2021 and be completed by December 2021. A copy of the PRISMA-P checklist is available in the online supplementary materials file 1. The final review will be guided by the Cochrane Handbook for Systematic Reviews of Interventions[34] and reported in accordance with the new Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 statement.[35]

Eligibility criteria

The PICO (population, intervention, comparison, outcomes) required by the PRISMA format are as follows:

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Population

Inclusion criteria are randomised and non-randomised controlled trials, noncontrolled trials, or observational studies with patients of any age with type 1 diabetes, type 2 diabetes, or gestational diabetes under the care of a primary care provider, including those co-managed with an endocrinologist. Included studies will be grouped according to the type of diabetes assessed (type 1, type 2, both type 1 and 2, and gestational diabetes). We will exclude studies which were primarily focused on participants who were critically ill, hospitalized or comprised of patients using closed loop systems.

Intervention

We will evaluate primary studies that report outcomes of the use of CGM or FGM in patients with diabetes in the primary care setting.

Comparison

The review seeks to compare the outcomes of the included studies on CGM/FGM with the practice of using HbA1c with or without self-monitoring of blood glucose levels in persons with diabetes in primary care practices.

Outcomes

Primary outcome

We have defined the primary outcome as the meeting of the ADA goals for glycaemic control in most patients of a HbA1c of ≤7% (≤53 mmol/mol, ≤154 mg/dL, ≤8.6 mmol/L).

Secondary outcomes

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We have included secondary outcomes of:

- change from baseline HbA1c,
- frequency of hypoglycaemia (≤3.8 mmol/L, or ≤68 mg/dL) and severe hypoglycaemia (≤ 3.0 mmol/L, or ≤54 mg/dL),
- hospitalization,
- time in target range (TIR),
- time above target range (TAR), and
- time below target range (TBR).[36]

For the secondary outcomes of TIR, TAR, and TBR, we will evaluate change from baseline if studies are unable to compare CGM/FGM intervention to control. For target ranges, we used a recent consensus report for patients with type 1 diabetes produced by the joint effort of multiple well recognized organizations that defined the target range of 3.9-10.0 mmol/L (70.2-180 mg/dL).[37] The goal of TIR > 70% for both patients with type 1 and type 2 diabetes (>50% for "older/high risk" individuals defined as cognitive deficits, renal disease, joint disease, osteoporosis, fracture, cardiovascular disease, and/or those requiring assisted care) was defined by a different recent consensus report.[38] For the subset of pregnant patients with diabetes, we will use the more stringent ADA guidelines.[39] If data exists, we will report on glycaemic variability, patient-oriented clinically significant outcomes of patient satisfaction and diabetes related complications such as major cardiovascular events, neuropathy, nephropathy, and retinopathy.

Type of studies

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We will include primary, peer reviewed studies where the abstract or manuscript is available in the English language that evaluate the use of CGM or FGM in a primary care clinic regardless of the design of the study, with a minimum of 20 participants. We will exclude reviews, editorials, case reports, conference abstracts and comments.

Information sources

We will include the electronic databases PubMed, MEDLINE, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science in the search for literature. All databases will be searched from their inception to February 2021. A full search update will be performed ahead of manuscript submission.

Search strategy

A comprehensive, systematic search strategy was constructed by a medical librarian (LÖ) and reviewed by the subject specialists (AK, RDG & JK). Search terms were defined based on the PICO (population, intervention, comparator, and outcomes) and the study inclusion and exclusion criteria as outlined above. PubMed and PubMed's MeSH was used in the pre-search phase in December-January 2021 to systematically identify and refine relevant search term variations and search techniques. The search strategy developed in PubMed will be adopted and applied in all selected databases without any geographical or publication year restrictions. We will use the "Text Word" field (or similar) for the best possible information retrieval. Hand screening of the references lists of the included studies will also be conducted.

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The search strategy developed through pre-searches in PubMed can be found in the online supplementary material file 2. A search log with notes, results, and details for the searches in all included databases in addition to a PRISMA 2020 flow-diagram outlining the details from the search and screening process will be appended to the final review. A search update in all databases will be conducted before finalizing the manuscript to ensure the inclusion of relevant studies published after the search.

Study records

Data management

We will upload all records identified in the database search to Covidence systematic review software[40] for automatic de-duplication and blinded screening, conflict resolving, study selection, and data extraction. The screening and extraction process in Covidence is blinded. Studies identified to be included in the review will be exported to EndNote, a citation management tool.

Selection process

The selection process in Covidence will follow the PRISMA workflow. Two independent reviewers (JK & RDG) will initially screen the title and abstracts of all identified papers based on the pre-set inclusion and exclusion criteria. A third reviewer (AK) will resolve eventual conflicts identified by the software. Full text of the papers identified as potentially eligible in the title and abstract screening will be reviewed independently by JK and AK. Reason for exclusion will be documented based on the pre-set study exclusion criteria imported to Covidence. RDG will resolve eventual conflicting exclusion reasons or paper

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inclusion/exclusion with the help of the software. Only the studies fulfilling the inclusion criteria will be fully screened for data extraction. The result from the screening and selection process will be documented in a PRISMA flow-diagram.

Data collection process

Two independent reviewers (JK & AK) will extract data from the final studies identified eligible to be included in the review. The Covidence Extraction module will be used to match the objectives of this study. Data to be extracted will include basic characteristics of the publication such as authors, year of publication, journal, and country. Additionally, the study design, population, duration and CGM usage duration will be taken into consideration. The primary variables to be collected will be age; type of diabetes (type 1, type 2 or gestational); the use of CGM, FGM or both; episodes of hypoglycaemia as defined above in the Outcomes section of the PICO statement; the number of insulin injections given per day (0, 1 or multiple); hospitalization for complications of either diabetes or diabetes-related treatment; patient satisfaction scores and physician satisfaction scores. Quantitative data to be extracted include the weighted mean difference of change from baseline HbA1c, time in (TIR), above (TAR), and below (TBR) the goal range after CGM/FGM is utilized. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be extracted.

Eventual discrepancies between the reviewers will be addressed with a third reviewer (RDG) and discussed until consensus is reached. Authors of the included papers will be contacted by e-mail in-case additional data is needed.

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Risk of bias in individual studies

Two independent reviewers (JK & RDG) will evaluate the risk of bias and the quality of evidence of all included studies using the National Heart, Lung, and Blood Institute (NHLBI) quality assessment tools.[41] The reviewers will critically assess each study for its methodology to determine if any attrition, publication, outcome, or conclusion biases are present. An appraisal of the quality of evidence for each of the outcomes will be given using the GRADE criteria.[42] Eventual disagreements, unusual patterns or discrepancies in the assessment will be reviewed and discussed together with a third reviewer (AK) until consensus is reached. The existence of systematic differences in baseline characteristics between the groups compared in a study will be addressed in the discussion at the end of the study.

Data synthesis

In a comprehensive table, we will summarize the PICO characteristics of each study including all relevant interventions and outcomes. Thereafter we will determine which studies have similar characteristics and outcomes, which will allow us to group them and compare their relevant findings. We anticipate these subgroups to include type 1 diabetes, type 2 diabetes without insulin therapy, type 2 diabetes with only long-acting insulin therapy, type 2 diabetes with multiple daily insulin injections and gestational diabetes. Given our review will be inclusive of many study designs, we may group data according to their respective study design. Quantitative data to be reported for each study will include the mean HbA1c at study end of the intervention group (CGM/FGM) compared to control. We will then take the mean differences with the standard errors of the means, calculate the 95% CIs, and graphically represent our

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results. Weighted mean differences of change from baseline HbA1c, TIR, TAR, and TBR after CGM/FGM intervention will similarly be analyzed. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be reported with their respective 95% CIs, along with qualitative data on patient and physician satisfaction. If any deviation from protocol plans is necessary, modifications to planned comparisons between studies will be made and noted. A narrative synthesis will be formulated for each subgroup of persons with diabetes, and we will give an appraisal of the utility of the addition of CGM or FGM, analyzing which subgroups appear to benefit most from CGM/FGM.

Strength of evidence

We will use the GRADE (Grading of Recommendations Assessment, Development and Evaluation)[42] tool to determine the strength of evidence, including publication bias, indirectness, inconsistencies, imprecision, and risk of bias. Two independent reviewers (JK & AK) will evaluate the quality of each study, and a third independent reviewer (RDG) will resolve any potential conflict.

Patient and public involvement: No patient or members of the public will be included in the performance of this study.

Discussion

This systematic review will be conducted to compare the benefits of adding CGM/FGM to usual care in the monitoring and management of diabetes mellitus in primary care. This technology has

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shown promise among persons with type 1 diabetes, and a systematic review and meta-analysis by Park and Le in 2018 demonstrated that CGM can have a modest improvement in HbA1c in type 2 diabetes.[6] However, it is unclear whether or not these benefits would be realized in the primary care setting where many persons with either type 1 or type 2 diabetes seek medical care. This review will allow us to determine whether the addition of CGM/FGM will lead to better outcomes in achieving glycaemic control while avoiding frequent hypoglycaemia, both of which have been strongly associated with reduced morbidity and mortality.

Ethics and dissemination: Considering the study design of systematic reviews, ethics approval is not needed. We intend to publish the review in a peer-reviewed journal where the results will be available in an open access format.

Authors' contributions: AK is the primary author for the study. The study design and draft of the protocol was conducted by AK, JK, RDG and LÖ. JK is the corresponding author. LÖ developed the search strategy and provided feedback on the study protocol design. The final manuscript was approved by all authors.

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Competing interest: None declared

Patient consent for publication: Not required.

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Provenance and peer review: not commissioned; externally peer reviewed.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist

Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and hemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol.

Authors: Dr. Alexander Kieu, Dr. Romona Govender Devi, Ms. Linda Östlundh & Dr. Jeffrey King.

Manuscript submission date: 2021-01-XX

Section and topic	Item No	Checklist item	Item reported on page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	**1**
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	**N/A**
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	**2 and 4** (Submitted on Jan. 11, 2021. Receipt number: 229416)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	**1**
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	**7**
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	**N/A**
Support:			
Sources	5a	Indicate sources of financial or other support for the review	**7**
Sponsor	5b	Provide name for the review funder and/or sponsor	**7**
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	**7**
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	**2-4**

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Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	**4 -5**
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	**4-5**
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	**5**
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, Including planned limits, such that it could be repeated	**5 and supplementary materials file 2**
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	**5**
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) hrough each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	**6**
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	**6**
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources) any pre-planned data assumptions and simplifications	**6**
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	**4-5**
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	**6**
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	**6**
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned explorationof consistency (such as I^2, Kendall's τ)	**6**
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses,	

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		meta-regression)	**6**
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	**N/A**
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	**6**
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	**7**

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary materials file 2: preliminary search strategy

Source: PubMed
Search date: 2021-01-24
Search specifications: All search terms are searched in the search field “TextWord” and in MeSH (when available). A filter for English language is applied.
Result: 110 records

Preliminary search strategy:

(("primary healthcare"[Text Word] OR "primary health care"[Text Word] OR “primary health-care”[Text Word] OR "Primary Health Care"[Mesh] OR "primary care"[Text Word] OR "primary medical care"[Text Word] OR “family medicine” [Text Word] OR "Family Practice"[Mesh] OR "family practice"[Text Word] OR “family physician*”[Text Word] OR “family practitioner*”[Text Word] OR “family clinic*”[Text Word] OR “general practice” [Text Word] OR “general practitioner*”[Text Word] OR "General Practitioners"[Mesh] OR "general clinic*"[Text Word] OR "general physician*"[Text Word] OR “ambulatory care” [Text Word] OR “ambulatory health care”[Text Word] OR “ambulatory clinic*” [Text Word] OR "ambulatory physician*”[Text Word] OR “ambulatory practitioner*”[Text Word] OR “ambulatory health center*”[Text Word] OR "Ambulatory Care Facilities"[Mesh] OR "Prenatal Care"[Mesh] OR "prenatal care"[Text Word] OR "prenatal health care"[Text Word] OR “antenatal care”[Text Word] OR “pregnancy care” [Text Word] OR “maternity care” [Text Word] OR “antenatal health care”[Text Word] OR “pregnancy health care” [Text Word] OR “maternity health care” [Text Word]) AND ("CGM"[Text Word] OR "Continuous Glucose Monitor*"[Text Word] OR "flash glucose monitor*" [Text Word]))

BMJ Open

The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and haemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol

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Keywords: Continuous glucose monitoring; primary care; diabetes mellitus; glycaemic control; systematic review; protocol

Word Count: 2,688 words

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Abstract

Introduction:

Studies demonstrate that optimal glycaemic control reduces morbidity from diabetes mellitus but remains elusive in a significant portion of patients. Although research shows that continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) improves glycaemic control in selected subsets of patients with diabetes in specialty practices, we found no systematic reviews evaluating the use of CGM/FGM in primary care, where the majority of patients with diabetes are cared for.

This systematic review aims to answer the questions: "Compared to usual care of self-monitoring blood glucose and HbA1c, does the addition of CGM/FGM use in the primary care of patients with diabetes improve glycaemic control, decrease rates of hypoglycaemia, and improve patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

Methods and Analysis:

Aligning with the PRISMA-P guidelines, a search will be conducted in PubMed, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science. We will include studies investigating CGM/FGM use and reporting the primary outcome measure of haemoglobin A1c and secondary outcome measures of hypoglycaemia, time in range (TIR), time below range (TBR), time above range (TAR), and patient/staff satisfaction. We will examine which patient populations appear to benefit from CGM/FGM. Three independent researchers will

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use the Covidence systematic review software for blinded screening and study selection. The National Heart, Lung, and Blood Institute (NHLBI) quality assessment tool and Grading of Recommendations Assessment, Development and Evaluation (GRADE) will be used to assess the risk of bias and quality of evidence.

Ethics and dissemination:

The systematic review methodology does not require ethics approval due to the nature of the study design. Study findings will be publicly available to a wide readership across disciplines and will be published in a peer-reviewed journal.

Prospero registration number: CRD42021229416

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Strengths and limitations of this study:

- To our knowledge, this will be the first systematic review to analyze the clinical benefits of CGM and FGM in the primary care setting.
- The evidence is located through a comprehensive and systematic search in seven biomedical and nursing databases.
- The Covidence systematic review software will be used for blinded screening, conflict resolving, extraction and quality assessment by three independent reviewers.
- NHLBI and GRADE will be used to assess the risk of bias and quality of evidence.
- Limitations include an English language bias and exclusion of grey, non-peer reviewed studies.

Introduction:

Diabetes mellitus continues to be a major contributor to morbidity and mortality worldwide. Despite considerable expenditures of precious healthcare funds, outcomes continue to be quite poor, even in resource-rich countries with developed healthcare systems. Diabetes and impaired fasting glucose are strongly associated with a large increase in cardiovascular and all-cause mortality.[1,2] Optimal glycaemic control has been shown in numerous studies to reduce the risk of morbidity,[3,4] but it has proved to be elusive in a significant portion of the patient population suffering from diabetes.[5]

Continuous glucose monitoring (CGM) and flash glucose monitoring (FGM) are technologies developed to add further precision to the monitoring and management of diabetes. Both CGM and FGM involve the use of subcutaneous sensors that sample interstitial glucose levels in the

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patient. In CGM, the sensor automatically sends glucose values to a receiver as often as every 5 minutes,[6] and it can be used to alert the patient when the glucose trend is expected to exceed set limits, i.e., result in hypoglycaemia or hyperglycaemia. Fingerstick calibration is required at least daily with CGM.[7] FGM, also called intermittently scanned CGM (isCGM), uses similar approaches, but values are measured on demand, i.e., the patient uses their smartphone or other device to interrogate the sensor to measure the current interstitial glucose level, rather than having continuous measurements made. FGM does not require fingerstick calibration, as calibration is performed at the factory.[8]

Numerous studies and reviews have evaluated the use of CGM and FGM in caring for a selected subset of patients with diabetes, and they have shown some clinical benefit with regards to the secondary endpoint of goal directed glycaemic control.[9] This benefit has been demonstrated in both patients with either type 1[10-13] or type 2 diabetes.[14,15] There is also evidence that the use of CGM in the management of pregnant patients with type 1 diabetes results in improved neonatal outcomes,[16] and CGM may improve the glycaemic control of adolescents with type 1 diabetes.[17] Although there is no recognized standard for what constitutes good glycaemic control, and most guidelines are now encouraging individualized haemoglobin A1c (HbA1c) goals based on patients' age, comorbidities, and life expectancy, for the purposes of this review we will use the American Diabetes Association (ADA) guidelines 2019 for a goal A1c in most patients of less than 7% (53 mmol/mol, 154 mg/dL, 8.6 mmol/L).[18]

A major limiting factor in achieving glucose targets with current standard practice is hypoglycaemia with intensification of medical management.[19] CGM/FGM have been shown to

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reduce episodes of both daytime and night-time hypoglycaemia.[15,20,21] Interestingly, a systematic review found that hypoglycaemia was very common in older adults treated with glucose-lowering agents, occurring in 28-65% of participants in the included studies, most of whom (80-100%) were asymptomatic.[22] Hypoglycaemia in this older patient population is associated with worsening cognitive impairment, dementia,[23] falls, fractures, and cardiovascular complications, including death.[24] Finally, recurrent hypoglycaemia can lead to impaired awareness of subsequent hypoglycaemic episodes.[25] Although the use of CGM/FGM was not found in a systematic review to lead to restoration of awareness of hypoglycaemia, the review found that CGM can reduce rates of severe hypoglycaemia without worsening glycaemic control.[26]

Studies have highlighted that only consistent use of CGM yields improved clinical outcomes,[27-29] which suggests patient satisfaction is clinically relevant. Although cutaneous complications with the use of CGM or FGM occur, less pain or discomfort was reported by most users of CGM/FGM than with the use of self-monitoring blood glucoses to test capillary blood glucose[30] resulting in a high overall level of satisfaction with CGM.[31]

Surveys have shown that there are an insufficient number of endocrinologists to care for the large population of patients with diabetes, and in fact, the US primary care workforce cares for 85% of patients with diabetes according to data from a 2014 survey.[32] However, although numerous studies have evaluated the use of CGM/FGM in specialty settings, our search showed there are no published systematic reviews for evaluating the use of CGM in the primary care setting. Thus, the purpose of this systematic review is to determine the benefits of CGM/FGM in the primary care of diabetes mellitus.

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Objectives:

This systematic review aims to answer the questions: "Compared to usual care of self-monitoring blood glucose and HbA1c, does the addition of CGM/FGM use in the primary care of patients with diabetes improve glycaemic control, decrease rates of hypoglycaemia, and improve patient and physician satisfaction?" and if so, "What subgroups of primary care patients with diabetes are most likely to benefit?"

Methods and analysis:

This protocol follows the PRISMA-P guidelines (Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocol)[33] and registered in the international prospective register of systematic reviews, PROSPERO: registration number CRD42021229416. Eventual amendments to this protocol will be recorded in the PROSPERO registration. It is anticipated that the study will commence in February 2021 and be completed by December 2021. A copy of the PRISMA-P checklist is available in the online supplementary materials file 1. The final review will be guided by the Cochrane Handbook for Systematic Reviews of Interventions[34] and reported in accordance with the new Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 statement.[35]

Eligibility criteria

The PICO (population, intervention, comparison, outcomes) required by the PRISMA format are as follows:

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Population

Inclusion criteria are randomised and non-randomised controlled trials, noncontrolled trials, or observational studies with patients of any age with type 1 diabetes, type 2 diabetes, or gestational diabetes under the care of a primary care provider, including those co-managed with an endocrinologist. Included studies will be grouped according to the type of diabetes assessed (type 1, type 2, both type 1 and 2, and gestational diabetes). We will exclude studies which were primarily focused on participants who were critically ill, hospitalized or comprised of patients using closed loop systems.

Intervention

We will evaluate primary studies that report outcomes of the use of CGM or FGM in patients with diabetes in the primary care setting.

Comparison

The review seeks to compare the outcomes of the included studies on CGM/FGM with the practice of using HbA1c with or without self-monitoring of blood glucose levels in persons with diabetes in primary care practices.

Outcomes

Primary outcome

We have defined the primary outcome as the meeting of the ADA goals for glycaemic control in most patients of a HbA1c of ≤7% (≤53 mmol/mol, ≤154 mg/dL, ≤8.6 mmol/L).

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Secondary outcomes

We have included secondary outcomes of:

- change from baseline HbA1c,
- frequency of hypoglycaemia (≤3.8 mmol/L, or ≤68 mg/dL) and severe hypoglycaemia (≤ 3.0 mmol/L, or ≤54 mg/dL),
- hospitalization,
- time in target range (TIR),
- time above target range (TAR), and
- time below target range (TBR).[36]

From our preliminary literature review, some studies used both SMBG and CGM/FGM in the control arm (but making clinical decisions based on SMBG) in order to compare change in TIR, TAR and TBR from baseline to study end to the intervention arm of CGM. However, for studies only evaluating change in TIR, TAR, and TBR in the intervention arm without a control, we will assess the percentage change from baseline to study end in these outcomes.

For target ranges, we used a recent consensus report for patients with type 1 diabetes produced by the joint effort of multiple well recognized organizations that defined the target range of 3.9-10.0 mmol/L (70.2-180 mg/dL).[37] The goal of TIR > 70% for both patients with type 1 and type 2 diabetes (>50% for "older/high risk" individuals defined as cognitive deficits, renal disease, joint disease, osteoporosis, fracture, cardiovascular disease, and/or those requiring assisted care) was defined by a different recent consensus report.[38] For the subset of pregnant patients with diabetes, we will use the more stringent ADA guidelines.[39] If data exists, we will report on glycaemic variability, patient-oriented clinically significant

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outcomes of patient satisfaction and diabetes related complications such as major cardiovascular events, neuropathy, nephropathy, and retinopathy.

Type of studies

We will include primary, peer reviewed studies where the abstract or manuscript is available in the English language that evaluate the use of CGM or FGM in a primary care clinic regardless of the design of the study, with a minimum of 20 participants. We will exclude reviews, editorials, case reports, conference abstracts and comments.

Information sources

We will include the electronic databases PubMed, MEDLINE, EMBASE, Scopus, CINAHL, Cochrane Central Register of Controlled Trials and Web of Science in the search for literature. All databases will be searched from their inception to February 2021. A full search update will be performed ahead of manuscript submission.

Search strategy

A comprehensive, systematic search strategy was constructed by a medical librarian (LÖ) and reviewed by the subject specialists (AK, RDG & JK). Search terms were defined based on the PICO (population, intervention, comparator, and outcomes) and the study inclusion and exclusion criteria as outlined above. PubMed and PubMed's MeSH was used in the pre-search phase in December-January 2021 to systematically identify and refine relevant search term variations and search techniques. The search strategy developed in PubMed will be adopted and applied in all selected databases without any geographical or publication year restrictions. We

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will use the "Text Word" field (or similar) for the best possible information retrieval. Hand screening of the references lists of the included studies will also be conducted.

The search strategy developed through pre-searches in PubMed can be found in the online supplementary material file 2. A search log with notes, results, and details for the searches in all included databases in addition to a PRISMA 2020 flow-diagram outlining the details from the search and screening process will be appended to the final review. A search update in all databases will be conducted before finalizing the manuscript to ensure the inclusion of relevant studies published after the search.

Study records

Data management

We will upload all records identified in the database search to Covidence systematic review software[40] for automatic de-duplication and blinded screening, conflict resolving, study selection, and data extraction. Studies identified to be included in the review will be exported to EndNote, a citation management tool.

Selection process

The selection process in Covidence will follow the PRISMA workflow. Two independent reviewers (JK & RDG) will initially screen the title and abstracts of all identified papers based on the pre-set inclusion and exclusion criteria. A third reviewer (AK) will resolve eventual conflicts identified by the software. Full text of the papers identified as potentially eligible in the title and abstract screening will be reviewed independently by JK and AK.

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Reason for exclusion will be documented based on the pre-set study exclusion criteria imported to Covidence. RDG will resolve eventual conflicting exclusion reasons or paper inclusion/exclusion with the help of the software. Only the studies fulfilling the inclusion criteria will be fully screened for data extraction. The result from the screening and selection process will be documented in a PRISMA flow-diagram.

Data collection process

Two independent reviewers (JK & AK) will extract data from the final studies identified eligible to be included in the review. The Covidence Extraction module will be used to match the objectives of this study. Data to be extracted will include basic characteristics of the publication such as authors, year of publication, journal, and country. Additionally, the study design, population, duration and CGM usage duration will be taken into consideration. The primary variables to be collected will be age; type of diabetes (type 1, type 2 or gestational); the use of CGM, FGM or both; episodes of hypoglycaemia as defined above in the Outcomes section of the PICO statement; the number of insulin injections given per day (0, 1 or multiple); hospitalization for complications of either diabetes or diabetes-related treatment; patient satisfaction scores and physician satisfaction scores. Quantitative data to be extracted include the weighted mean difference of change from baseline HbA1c, time in (TIR), above (TAR), and below (TBR) the goal range after CGM/FGM is utilized. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be extracted.

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Eventual discrepancies between the reviewers will be addressed with a third reviewer (RDG) and discussed until consensus is reached. Authors of the included papers will be contacted by e-mail in-case additional data is needed.

Risk of bias in individual studies

Two independent reviewers (JK & RDG) will evaluate the risk of bias and the quality of evidence of all included studies using the National Heart, Lung, and Blood Institute (NHLBI) quality assessment tools.[41] The reviewers will critically assess each study for its methodology to determine if any attrition, publication, outcome, or conclusion biases are present. An appraisal of the quality of evidence for each of the outcomes will be given using the GRADE criteria.[42] Eventual disagreements, unusual patterns or discrepancies in the assessment will be reviewed and discussed together with a third reviewer (AK) until consensus is reached. The existence of systematic differences in baseline characteristics between the groups compared in a study will be addressed in the discussion at the end of the study.

Data synthesis

In a comprehensive table, we will summarize the PICO characteristics of each study including all relevant interventions and outcomes. Thereafter we will determine which studies have similar characteristics and outcomes, which will allow us to group them and compare their relevant findings. We anticipate these subgroups to include type 1 diabetes, type 2 diabetes without insulin therapy, type 2 diabetes with only long-acting insulin therapy, type 2 diabetes with multiple daily insulin injections and gestational diabetes. Given our review will be inclusive of many study designs, we may group data according to their respective study design.

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Quantitative data to be reported for each study will include the mean HbA1c at study end of the intervention group (CGM/FGM) compared to control. We will then take the mean differences with the standard errors of the means, calculate the 95% CIs, and graphically represent our results. Weighted mean differences of change from baseline HbA1c, TIR, TAR, and TBR after CGM/FGM intervention will similarly be analyzed. Additionally, data on the relative risks of hospitalizations, hypoglycaemia and severe hypoglycaemia will be reported with their respective 95% CIs, along with qualitative data on patient and physician satisfaction. If any deviation from protocol plans is necessary, modifications to planned comparisons between studies will be made and noted. A narrative synthesis will be formulated for each subgroup of persons with diabetes, and we will give an appraisal of the utility of the addition of CGM or FGM, analyzing which subgroups appear to benefit most from CGM/FGM.

If there are sufficient RCTs with low variability among them, a meta-analysis will be attempted comparing CGM/FGM to SMBG. A sensitivity analysis will be conducted by excluding studies with a high risk of bias, which will confirm the robustness of our results. We will assess heterogeneity using the I^2 statistic, where ≥50% is considered heterogenous.[42,43] If heterogeneity is determined, a meta-regression analysis will be performed to evaluate whether baseline HbA1c, duration of CGM/FGM use, and study design influence CGM/FGM's effects on HbA1c. Egger's regression test will be used to detect publication bias.[44]

Strength of evidence

We will use the GRADE (Grading of Recommendations Assessment, Development and Evaluation)[45] tool to determine the strength of evidence, including publication bias, indirectness,

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inconsistencies, imprecision, and risk of bias. Two independent reviewers (JK & AK) will evaluate the quality of each study, and a third independent reviewer (RDG) will resolve any potential conflict.

Patient and public involvement: No patient or members of the public will be included in the performance of this study.

Discussion

This systematic review will be conducted to compare the benefits of adding CGM/FGM to usual care in the monitoring and management of diabetes mellitus in primary care. This technology has shown promise among persons with type 1 diabetes, and a systematic review and meta-analysis by Park and Le in 2018 demonstrated that CGM can have a modest improvement in HbA1c in type 2 diabetes.[6] However, it is unclear whether these benefits would be realized in the primary care setting where many persons with either type 1 or type 2 diabetes seek medical care. This review will allow us to determine whether the addition of CGM/FGM will lead to better outcomes in achieving glycaemic control while avoiding frequent hypoglycaemia, both of which have been strongly associated with reduced morbidity and mortality.

Ethics and dissemination: Considering the study design of systematic reviews, ethics approval is not needed. We intend to publish the review in a peer-reviewed journal where the results will be available in an open access format.

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Authors' contributions: AK is the primary author for the study. The study design and draft of the protocol was conducted by AK, JK, RDG and LÖ. JK is the corresponding author. LÖ developed the search strategy and provided feedback on the study protocol design. The final manuscript was approved by all authors.

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Competing interest: None declared

Patient consent for publication: Not required.

Provenance and peer review: not commissioned; externally peer reviewed.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist

Title: The benefits of the addition of continuous or flash glucose monitoring versus standard practice utilizing self-monitored blood glucose and hemoglobin A1c in the primary care of diabetes mellitus: a systematic review protocol.

Authors: Dr. Alexander Kieu, Dr. Romona Govender Devi, Ms. Linda Östlundh & Dr. Jeffrey King.

Manuscript submission date: 2021-01-XX

Section and topic	Item No	Checklist item	Item reported on page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	**1**
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	**N/A**
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	**2 and 4** (Submitted on Jan. 11, 2021. Receipt number: 229416)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	**1**
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	**7**
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	**N/A**
Support:			
Sources	5a	Indicate sources of financial or other support for the review	**7**
Sponsor	5b	Provide name for the review funder and/or sponsor	**7**
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	**7**
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	**2-4**

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Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	**4 -5**
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	**4-5**
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	**5**
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, Including planned limits, such that it could be repeated	**5 and supplementary materials file 2**
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	**5**
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) hrough each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	**6**
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	**6**
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources) any pre-planned data assumptions and simplifications	**6**
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	**4-5**
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	**6**
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	**6**
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned explorationof consistency (such as I^2, Kendall's τ)	**6**
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses,	

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		meta-regression)	**6**
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	**N/A**
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	**6**
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	**7**

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary materials file 2: preliminary search strategy

Source: PubMed
Search date: 2021-01-24
Search specifications: All search terms are searched in the search field “TextWord” and in MeSH (when available). A filter for English language is applied.
Result: 110 records

Preliminary search strategy:

(("primary healthcare"[Text Word] OR "primary health care"[Text Word] OR “primary health-care”[Text Word] OR "Primary Health Care"[Mesh] OR "primary care"[Text Word] OR "primary medical care"[Text Word] OR “family medicine” [Text Word] OR "Family Practice"[Mesh] OR "family practice"[Text Word] OR “family physician*”[Text Word] OR “family practitioner*”[Text Word] OR “family clinic*”[Text Word] OR “general practice” [Text Word] OR “general practitioner*”[Text Word] OR "General Practitioners"[Mesh] OR "general clinic*"[Text Word] OR "general physician*"[Text Word] OR “ambulatory care” [Text Word] OR “ambulatory health care”[Text Word] OR “ambulatory clinic*” [Text Word] OR "ambulatory physician*”[Text Word] OR “ambulatory practitioner*”[Text Word] OR “ambulatory health center*”[Text Word] OR "Ambulatory Care Facilities"[Mesh] OR "Prenatal Care"[Mesh] OR "prenatal care"[Text Word] OR "prenatal health care"[Text Word] OR “antenatal care”[Text Word] OR “pregnancy care” [Text Word] OR “maternity care” [Text Word] OR “antenatal health care”[Text Word] OR “pregnancy health care” [Text Word] OR “maternity health care” [Text Word]) AND ("CGM"[Text Word] OR "Continuous Glucose Monitor*"[Text Word] OR "flash glucose monitor*" [Text Word]))